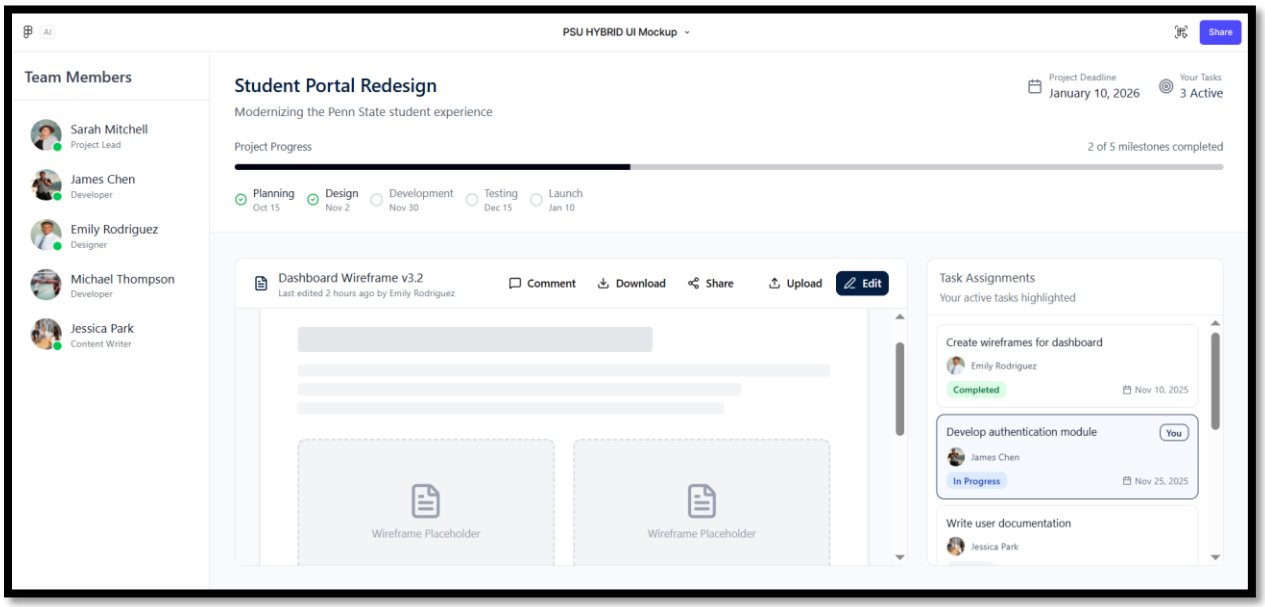
Systems Engineering: An Overview of Hybrid Work Unfairness and System Inefficacy, and a Proposed Software Design Solution.



Anna H. Jordan

John Alberta

Lina Y. Elhassan

EDSGN 100: Cornerstone Engineering Design
Fall 2025 Client-Driven Design Project

Section [009], [**Dr.Tianliang Yu**]

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PennState College of Engineering

ENGINEERING DESIGN AND INNOVATION

Executive Summary

This report addresses the needs of the hybrid staff at Penn State's School of Engineering Design and Innovation (SEDI) and proposes a solution based on findings.

1.0. Problem understanding:

As the complexity increases, more system-related problems arise. While hybrid work is beneficial, it adds another level of complexity to the traditional office.

To solve the problems of hybrid work and maintain its benefits, we must look at the structure of the whole system. This prevents the creation of a new problem in a different part of the system while allowing for more efficiency in problem-solving.

To understand the problems of any system, we must go through research related to its different aspects.

The first aspect is the ***sustainability of the system****. According to Penn State's definition of sustainability: "The simultaneous pursuit of human health and happiness. Environmental Quality and economic well-being for current and future generations." (Penn State sustainability, 2025)*

The second aspect was gathering ***interview results*** *from hybrid workers. Gathering information from hybrid workers is crucial to give us a holistic scope of the problems.*

The third component was the ***system overview****. This included the evaluation of system parts: Employer, Employees, Worksite, IT, and Families of workers.*

The final aspect was the ***stakeholder assessment****. It is important to highlight different stakeholders. Having detailed knowledge of the stakeholders helped us identify their pain points.*

2.0. First Design Iteration:

This iteration of the initial prototype process:

1. *Preliminary ideation: Rapid ideation for possible solutions to problems. This ideation process was conducted through questions generated from various stakeholder needs. This involved drawing diagrams to communicate our ideas. There were four focuses for this ideation:*

- *Constraint-focused ideation: Involved the addition of constraints to different problems, such as budget, size, and energy use.*
- *Stakeholder-focused ideation: Revolved only around stakeholders' needs using our research.*
- *Sustainability-focused: Concerned with the carbon footprint and following the Penn State definition of sustainability.*

2. *Preliminary evaluation of the screened concepts: Filtering the concepts and assigning the most efficient four concepts to carry to the next phase.*

3. *Concept selection: Two concepts were picked from the concept screening phase to continue to the second design iteration.*

3.0. Second Design Iteration:

Included the gathering interviews for the two selected concepts. The interviews were collected from hybrid workers to gain additional insight into the concepts.

- *After the second design iteration, only one concept was selected to move forward with.*

4.0. Final Design Description:

The final concept selected was carried through to the generation of the final design phase.

The final product design is a platform for employees to connect inside and outside the office. The software features include:

1. Common workspace

2. Contacts and ability to reach other employees

3. Tasks assigned

4. Project completion status

5. End of the day prompts.

Conclusion:

The product was named PSU Orbit; it was designed to increase fairness and productivity. PSU Orbit provides Penn State with control over data and privacy policy. Utilizing PSU Orbit is the best monetary investment in comparison to subscription-based platforms online.

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1.0 Problem Understanding

This project is concerned with Hybrid work. A Hybrid Worker is defined as “a person who works remotely or from home part of the time and on-site at a workplace (esp. a traditional office environment) the rest of the time” [1]. This means that employees spend some of their working hours in the office while having the ability or choice to work outside of the office. The scope of this project covers some of the problems of hybrid work and their potential solutions. Specifically focusing on the problems evolving Staff of Engineering Design and Innovation building (SEDI)

A hybrid system in general is a system that involves two components. Problems with hybrid work are mostly due to trade-offs, and this means that having one solution or one aspect added to a system may result in a new problem arising. For example: hiring more workers to help with the workload might overcrowd the office spaces. For this reason, we must consider the potential solutions while keeping in mind the stakeholders “individuals or groups who are affected by or have an interest in a particular project or systems” [2]. Knowing how the stakeholders relate to the system helps identify potential solutions without creating problems in other parts of the system.

Stakeholders for the SEDI include the employees and staff, Pennsylvania State University, Sponsors, and Students. These stakeholders will be explored later in this paper in more detail. The problems with hybrid work vary according to which stakeholders are affected. Some examples of those problems include unfairness, time management, loss of resources, stress, overtime, exhaustion, work-life balance, environmental pollution and impact, efficiency, communication difficulty, and many more.

1.1 Sustainability Overview

Sustainability not only comes from a lower carbon footprint but also the ability to continue it years from now. The United Nations has 17 goals for Sustainable Development to which the top five is, No Poverty, Zero Hunger, Good Health, well-being, quality education, and gender equality [3]. Hybrid work has been proven to improve workers work life balance, provides flexibility to make appointments, and allow for more family time. All these things will lead to better well-being and great health in employees. Penn State describes sustainability as, “Sustainability is the simultaneous pursuit of human health and happiness, environmental quality, and economic well-being for current and future generations” [4]. As shown before the action of hybrid work represents both Penn State's and the United Nations descriptions of sustainability.

Another side of sustainability is reducing environmental impact of a process or company. The biggest changes seen from hybrid work is a reduction in commutes to and from the office leading to lower carbon emissions. In these systems, you also see a reduction in electricity usage from the company; however, it can be argued that to net total energy consumption is the same as non-hybrid work because of the need for home office spaces. The reduction in power usage is not for the company's longevity due to less overhead but it's good for the environment due to the lower need or

energy which is mostly carbon based. This aligns with the president of Penn States statement, "...goal of achieving 100% greenhouse gas emissions reduction by 2035..." [5].

1.2 Interview Results

Interviews with hybrid workers were extremely helpful in identifying both challenges and benefits of working in a hybrid environment. These interviews were conducted over a wide range of workers, with people working both within and outside of Penn State University included. Responses from workers inside Penn State's College of Engineering Design and Innovation were also included [6] to better focus analysis on to the target user group. Overall, the interviews generally listed similar benefits, challenges, and complaints in hybrid work environments, regardless of the actual job or responsibilities.

When working from home, there were lots of benefits across the board that the interviewees mentioned. Working from home allows for increased flexibility on how the workday is timed and accomplished (e.g. starting the workday later to accomplish other tasks in the morning), as well as a space that allows for better focus and less distractions that come with an in-person worksite. Any time that is usually wasted on tasks unique to working in-person (e.g. commuting, getting ready, interactions with co-workers) is cut down. Overall, one of the major benefits was mentioned to be a better work-life balance and a more sustainable lifestyle compared to working fully in-person.

On the other hand, there were still challenges from working strictly remotely, showing the need for a hybrid work environment. When fully working remotely, the in-person connection with people like co-workers and managers is lost, and a lot of the benefits from that (e.g. networking, making friends) are also lost. Professional communication can also be more challenging, as the interviewees complained that sometimes people don't check their email or use digital communication tools (e.g. Microsoft Teams) often. One of the interviewees specifically mentioned that working remotely can be difficult early on in one's career, because there's a lot of "micro-interactions" that are important for networking and for building professional connections. Later on in one's career, these aren't necessarily as important, but early on, they can help greatly. Additionally, it can be hard to mimic an in-person work setup, and so often ergonomic (chairs, desks) or technological equipment (cameras, microphones) can be hard to obtain while working from home.

When working in-person, a lot of the issues that come with working remotely are fully solved (better communication, better connections, less distractions), however there are still challenges with working in-person. In general, there's a lower work-life balance as more time in the day is dedicated to work (commuting, meetings, etc.), and you have to fully be present at the office for the full work-day, instead of being able to fit in other tasks during downtime when you're working remotely.

Overall, there were a lot of challenges and benefits that came with both working remotely and working from home. Working from home brings better flexibility and focus, but less connections and ability to perform in-person tasks, while working in-person brings better collaboration and access to facilities, but less work-life balance and overall flexibility. Above all else, the interviewees really stressed the importance of flexibility and autonomy with how they conduct their work schedules, as different people work better in different environments. A hybrid work environment does seem to mitigate a lot of the challenges that are brought with working fully in-person versus fully remotely.

When asked if they would recommend hybrid work to a friend or family member, the SEDI interviewees responded, “overwhelmingly yes”, which really highlights the benefits of hybrid work, and the overall attitude modern workers have towards it.

1.3 System Overview

When we did the characterization for our mind-map, we found the most common connections were between different stakeholders affected by hybrid work. As we linked the different points, we noticed many of our stakeholders have shared interests, leading us to combine them for the map. Once we found those connections, we created a whiteboard into five main categories: Employee (Figure 1.3.1), Employer (Figure 1.3.2), Family (Figure 1.3.3), Office/Worksite (Figure 1.3.4), and IT (Figure 1.3.5). Within each of the main groups we discovered that we could create sub-groups, those groups being a benefit and a challenge experienced by each stakeholder. While looking over the mind-map we discovered some groups not covered so then moving forward we included them.

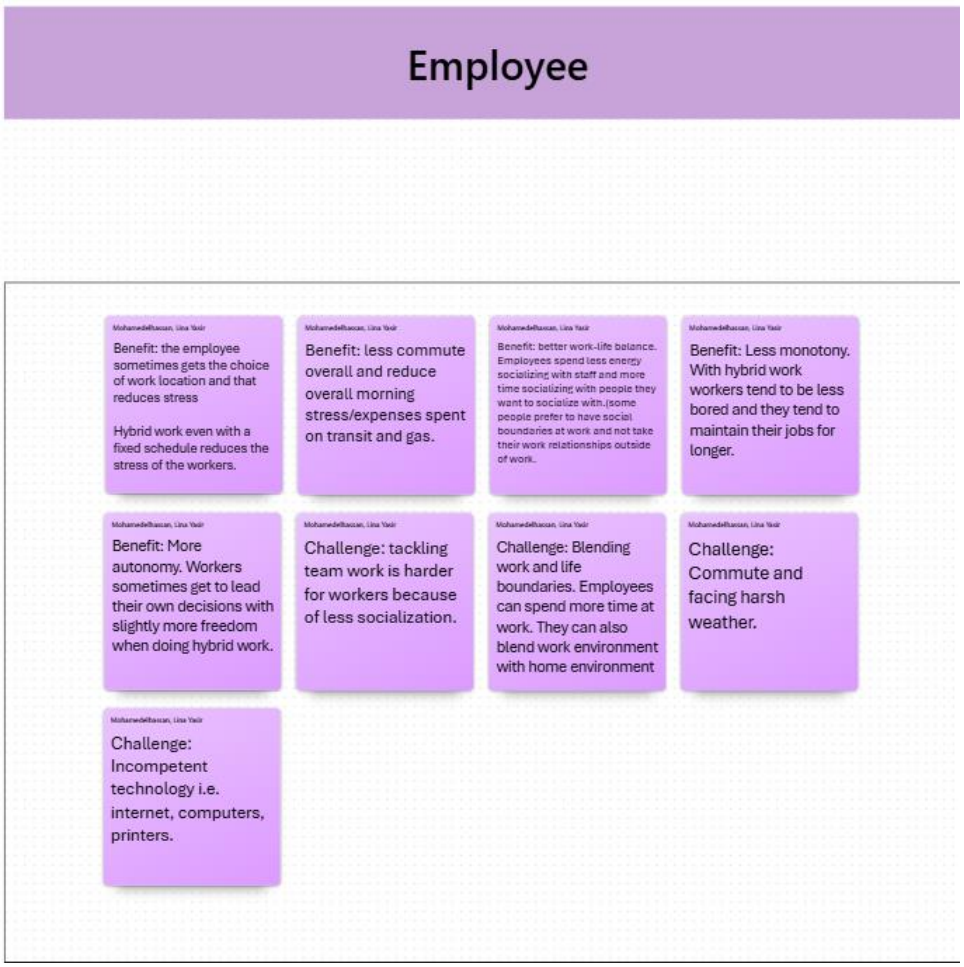

Figure 1.3.1: Mind map for Employee

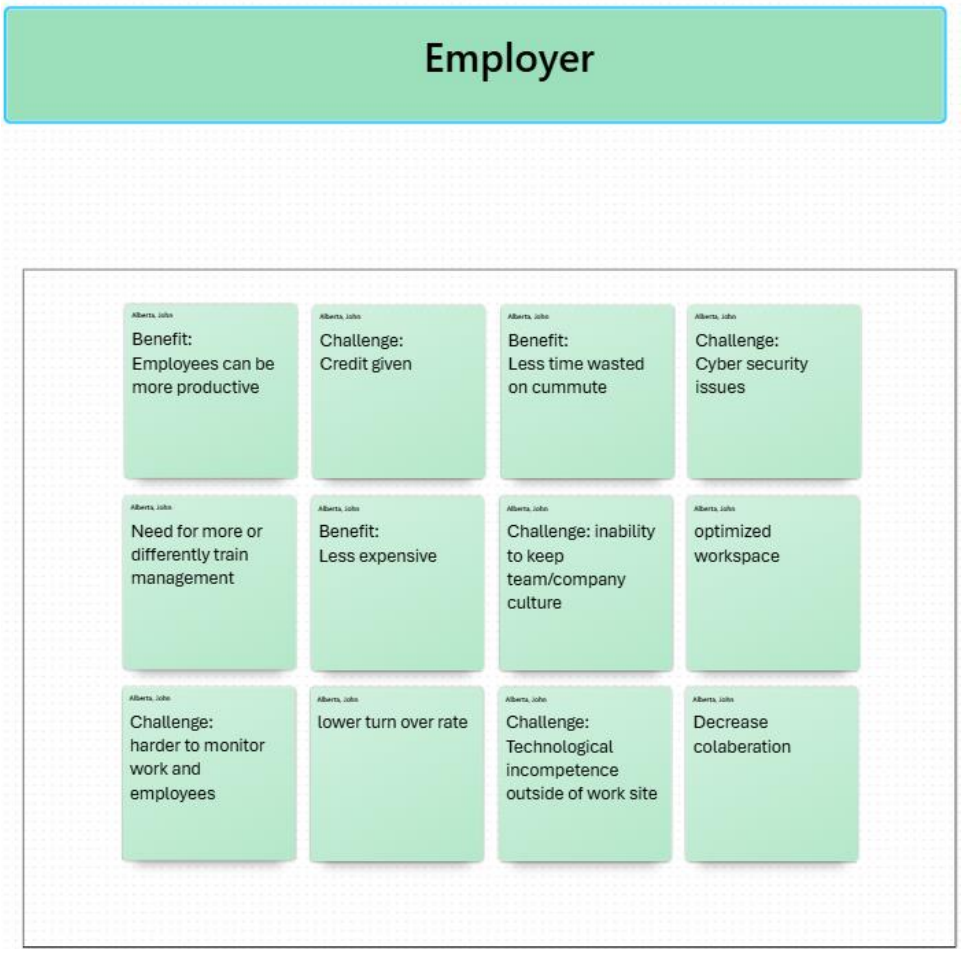

Figure 1.3.2: Mind map for Employer

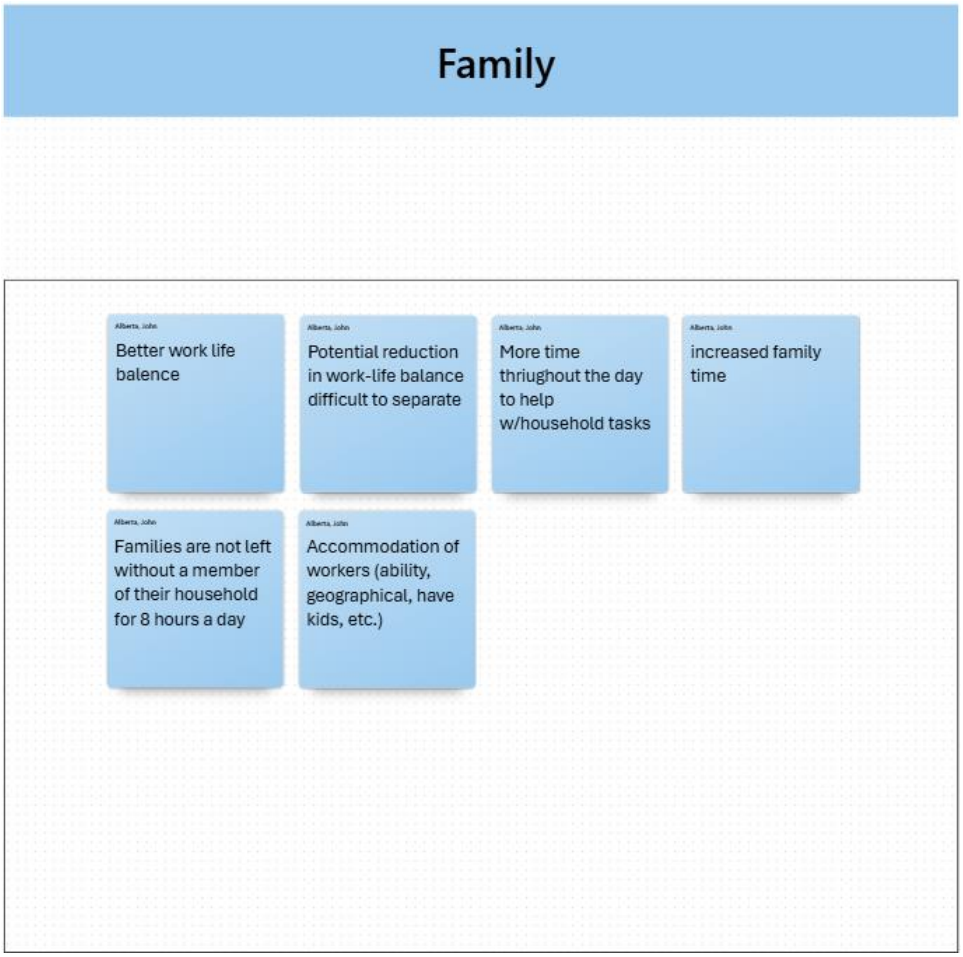

Figure 1.3.3: Mind map for Family

Figure 1.3.4: Mind map for Office/Worksite

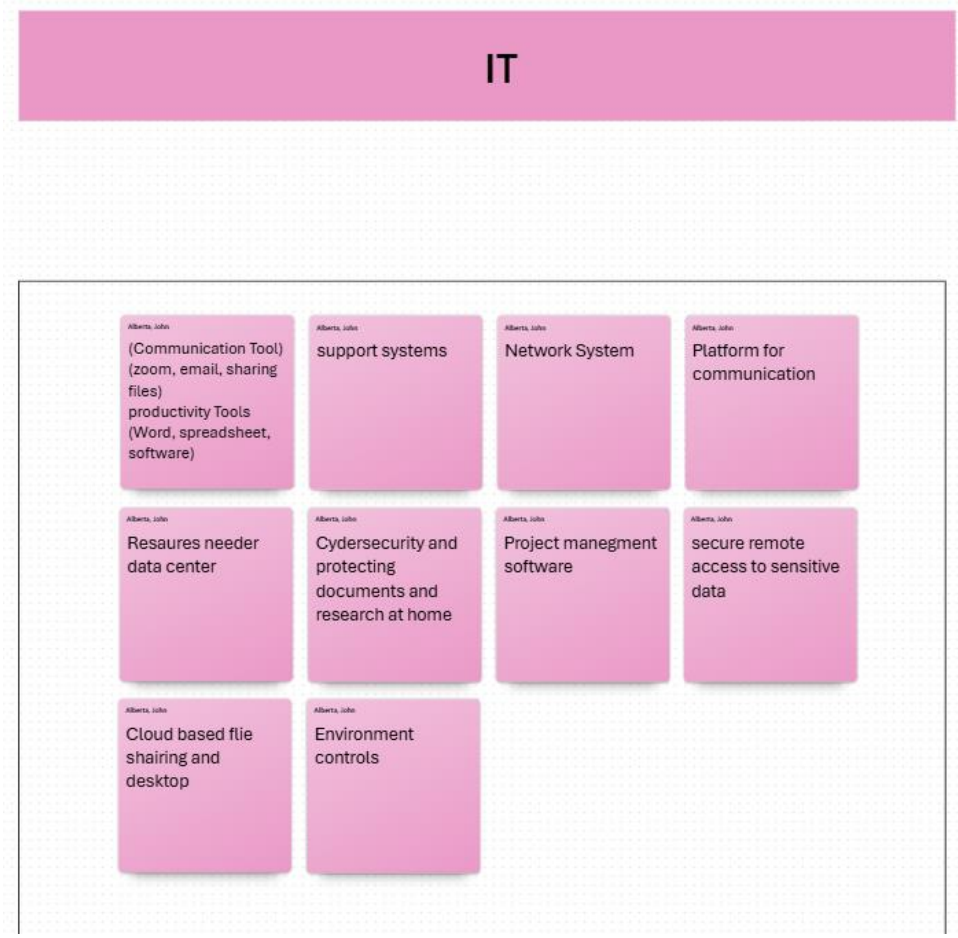

Figure 1.3.5: Mind map for IT

1.4 Stakeholder Assessment

As mentioned previously, stakeholders are mainly split into four groups:

Employees, Penn State University, Sponsors, and Students.

Those main groups would then diverge to show more stakeholders as in Table 1.

Table 1. the main stakeholders considered in the project: Hybrid work problems

1	Employees	Instructors	IT faculty	Management
2	Penn State	Environment	Old main	Alumni/Donors
3	Sponsors	Partners	Technology Vendors	Organizations
4	Students	Professors	Class sections	Degree progression

Those main stakeholders could then be categorized into Internal, Institution, External, and Organization with regard to the institution. This categorization helps with the introduction of pain points for Hybrid work.

For internal stakeholders:

Employees:

1. Work life balance
2. Time management
3. Communication
4. Workspace
5. Unfairness and disregard of efforts

Students:

1. Lectures and trainings
2. Need to use maker spaces with assistance and training

For external stakeholders:

Sponsors:

1. Communication inefficiency
2. Loss of resources

Partners:

1. Lack of communication
2. Disagreements

For Institution stakeholders:

Penn state:

1. Financial trouble

2. Environment pollution
3. Loss of resource
4. Wages

Alumni and donors:

3. School focusing on monetary value more than education

For Outside Organizations stakeholders:

Organization:

5. Training difficulties
6. Performance tracking difficulties

Pain points are a big factor in decision-making, but they also help in recognizing the needs associated with the stakeholders. Those needs could also be summarized according to the different stakeholders. The needs would address everything that helps the stakeholder complete their role effectively while being compensated fairly for their work. Providing each stakeholder with their needs would help the system function without complications.

Table 2. Needs of stakeholders.

Stakeholder	Needs
Employees	- Comfortable workspace - Comfortable wages - Flexibility - Work-life balance - Efficiently - Better communication
Students	- Communication - Meet with instructors - Trainings
Sponsors	- High quality performance - Good use of resources - Maintain school prestige
Partners	- Good communication - Reliability - Good work ethic
Penn state	- Reliable staff - Strong program - More donors, partners and sponsorships
Alumni	- Good education - Recognition - Strong school spirit

Organizations	- Reliable and strong educational programs to support or fund

2.0 First Design Iteration

2.1 Preliminary Ideation

The ideation process for this project started individually, and then the ideas were discussed through team collaboration. Our team started with preliminary ideation and slowly built up to adding constraints. The process consisted of sketching the ideas which helped with visual conceptualization.

The approach used for the preliminary ideation was the “How Might We” questions or “HMW” for short. Those questions are based on the Penn state HR107 Staff Flexible Work Arrangements Policy that includes a focus on:

- Promoting productivity
- Promoting work/life balance
- Reducing overall greenhouse gas emissions
- Keeping a robust university workspace or on-campus experience

This policy helps us to address certain points of interest and would help us propose the right “how may we” questions. Those questions were used for concept for ideation. Some of the most impactful HMW questions were:

- How might we, improve communication between team members?
- How might we, make employees feel more included?
- How may we, train new hires better and get them used to the work environment without creating unfairness?
- How might we help employees mentally “clock out” and create a healthy work-life balance?
- How might we, maintain outside of office efficiency while keeping employees happy?

Samples of the sketched preliminary ideas:

The first sample in figure 2.1.1 describes a system of task tracking and workflow sharing between teammates. This system helps with fair work distribution because of the clear boundaries set by the system and that helps employees keep up with the schedule. This would also maintain productivity because employees would see needs to be done on their screen.

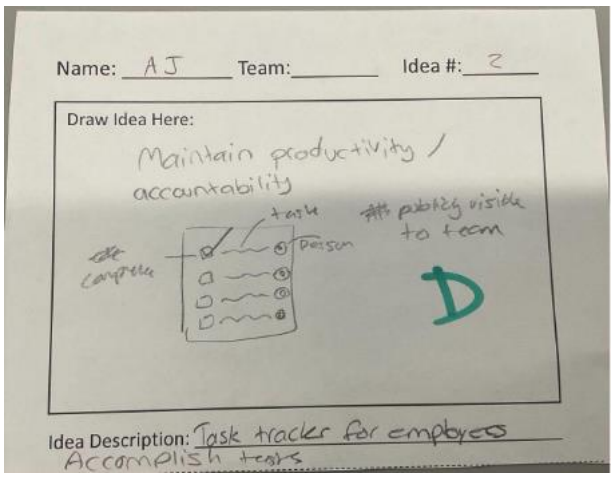

Figure 2.1.1. An illustration of the interface of the task tracker system for work projects and general tasks.

Another sample idea is the addition of a “system shutdown” concept which helps by reducing burnout. The idea is to provide a set time for when employees should stop working (for example 4pm). The software would send a notification to the user signalling the end of the workday and that the system would shut down. Unfortunately, this concept is very rigid and unrealistic therefore some improvements should be made. The improvements are replacing the complete system-shut down with either the employee reaching out to the manager asking for an extension or a questionnaire that takes the employee out of the “work-flow state”. The questionnaire may include questions such as:

1. What are you going to have for dinner?
2. What’s the weather like today?
3. Do you feel like going to bed?

Those questions are aimed at taking the employees’ focus away from work. The questionnaire system would be less rigid because it is easier to skip the questions and carry on with work. This system helps with the burn-out associated with overwork, which is a problem related to hybrid work.

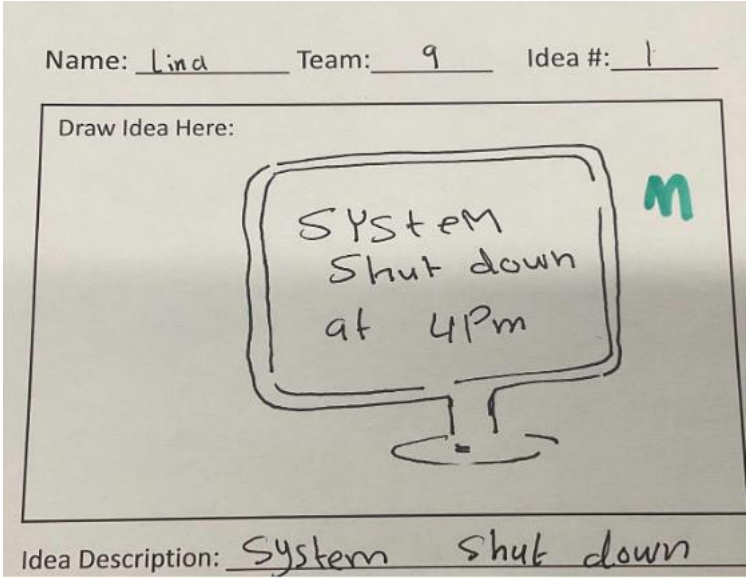

Figure 2.1.2. A sketch of the work system shutting down after the workday ends.

The third sample concept was the Remote Shared Workspaces, illustrated in Figure 2.1.3. The concept is having the system functioning as an office environment. This concept targets the difficulty of communication in hybrid work environments. The concept is to have the platform connecting all users together. This platform would show who is in the office and who is out (working from home). Employees in the office should also be using the same system, and this helps in making sure everyone feels included. The platform may include all the shared work documents and the project information and deadlines. Since all employees are using the software for their work (inside and outside of the office) this helps them receive and respond to communication alerts faster.

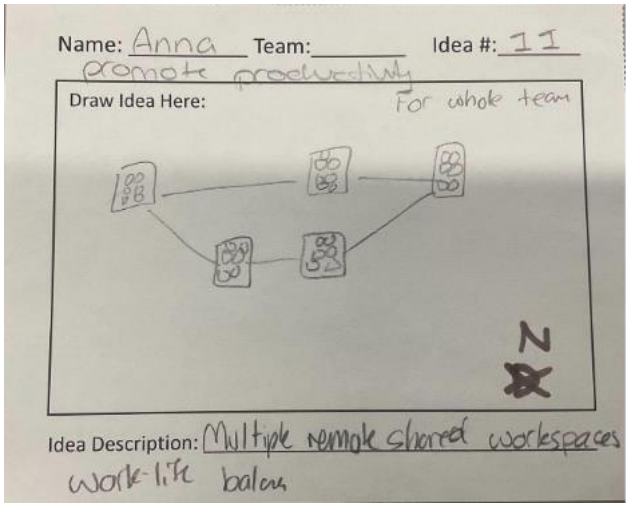

Figure 2.1.3. A sketch of shared online workspaces where tasks can be rotated and viewed outside and inside the office.

Another concept we thought of at our preliminary ideation phase was a physical board designed for the on-site workers. The board has a mechanical swipe function that once applied would show who is in the office at the time of swiping and whoever is working remote would swipe to the other side. The swiping mechanism is shown in Figure 2.1.4. This board functions as a displayed schedule for the week.

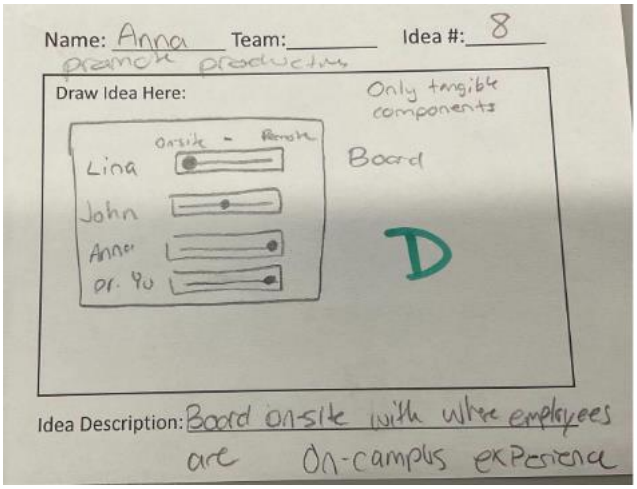

Figure 2.1.4. Illustration of the in-office board that shows who is working in the office vs. remote depending on which direction you swipe

The idea of the in-office board could be digitalized and made into a shared workspace system.

2.2 Constraint-focused Ideation

While ideating we went over eight different constraints all the way from cost to size of the product. The first two rounds we had to work with vastly different values, the first being $50 and the second $100 million dollars. Following this, we had to change the physical properties of the product first by fitting it into a lunch bag, then a tangible product preceded by an intangible product. We then thought of designing a product for a team or for an individual. Focusing on the cost is shown in Figure 2.2.1, which is based on the $50 limit. This idea was to create something for a more ergonomic office; this planned to help a major complaint of workers where it would be uncomfortable to share with someone else, hopefully leading to more productivity. Then, for Figure 2.2.2, we were constrained to fit it in a lunch bag. To combat poor communication between co-workers we would supply a company phone to all employees. Finaly Figure 2.2.3 is an intangible product with an integrated communication and work application. This allows for seamless communication while working promoting collaboration.

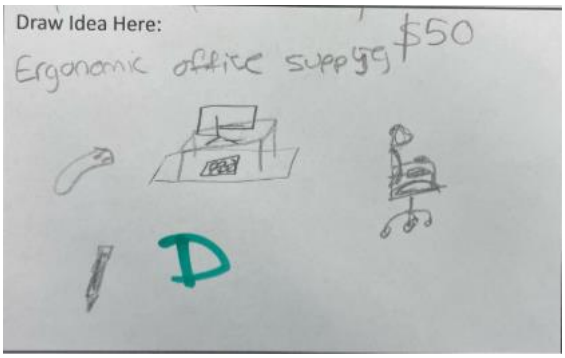

Figure 2.2.1. A sketch of different Ergonomic office supplies.

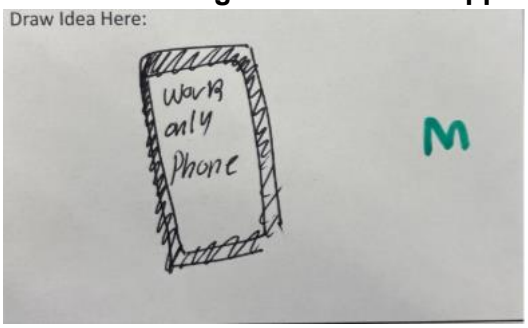

Figure 2.2.2. Representation of the work phone.

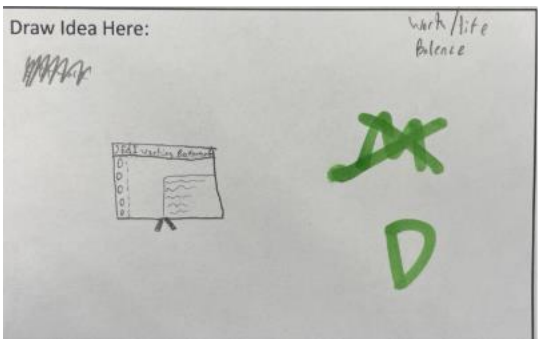

Figure 2.2.3. First sketch of the integrated communication system.

2.3 Stakeholder-focused Ideation

Another perspective for ideation was focusing on the stakeholders' needs and interests. For this form of ideation, we used our list of stakeholders and discussed what idea would benefit which of the stakeholders.

<u>Some samples and figures of the stakeholder ideation:</u>

The first example is the idea of hosting more work events regularly and this would specifically benefit Penn State and the SEDI as a staff. Penn State and SEDI are the main stakeholders for this concept, and they are benefiting by reducing stress and raising the company's spirit, which promotes productivity and lowers general turnover rates. This concept would help the bonding between hybrid workers. Those work events could occasionally include inviting sponsors to also target sponsorships as a stakeholder. Figure 2.3.1 illustrates some event-related activities that help employees' distress and get a break from work.

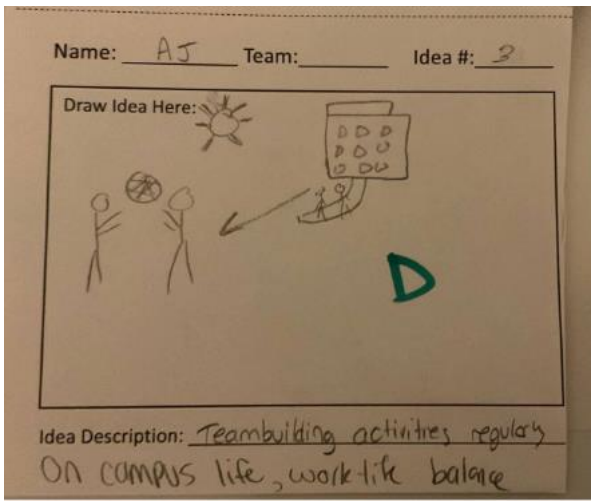

Figure 2.3.1. Team building activities that help employees create connections and help Penn State achieve a strong staff.

Sample two includes ideation related to the environment as a stakeholder. The idea is to change the inkjet on-campus printers to laser printers. While laser printers are relatively expensive and they use a lot of energy they are still less wasteful compared to inkjet printers. Inkjet printers have a high frequency for cartridge and ink changing due to their cleaning method, therefore it's more wasteful. Figure 2.3.2 shows the ideation of the concept.

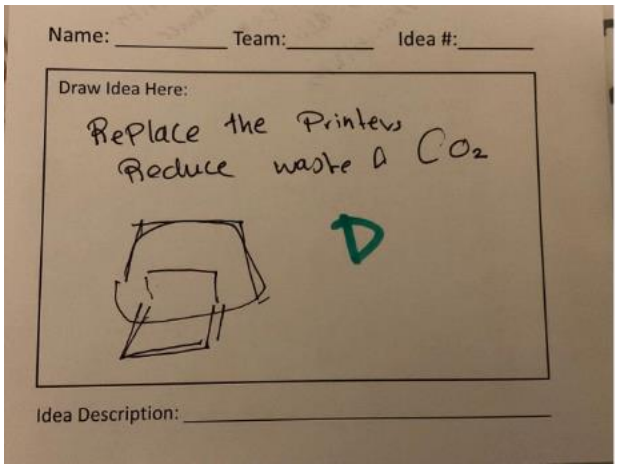

Figure 2.3.2. Illustrates the idea of replacing inkjet (wasteful) with a more environmentally sustainable one printer such as laser printers.

The third stakeholder focused idea was to maintain the employee's wellbeing, so the main stakeholder targeted were the employees. This mainly applies to employees in the office but could also be implemented into a system for those working remotely.

The main idea is to have a way to motivate workers to take breaks from work by having a reminder that is set up inside their work platform. This reminder will be shared with all the employees on or offsite. Figure 2.3.3 shows a demonstration of someone using their computer in the office and then getting a notification to go on break as their fellow employees leave for lunch.

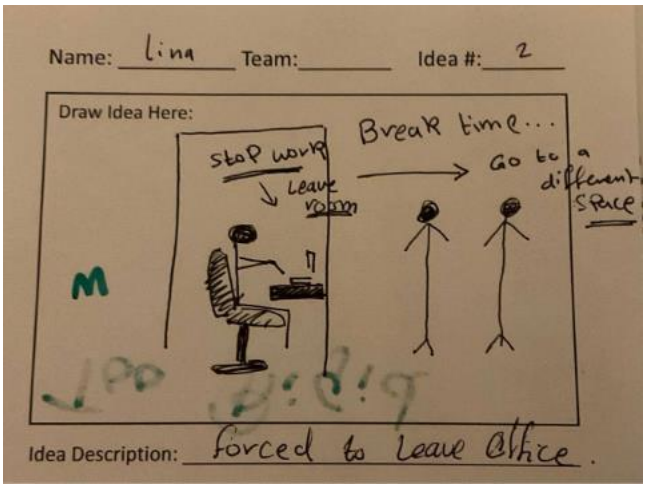

Figure 2.3.3. A worker getting a notification on their computer to stop working and get lunch with the other employees.

The fourth stakeholder focused example was centred around Donors, sponsors and Old main. The concept is in subsidizing the office equipment which would help maximize worker productivity due to the improved working conditions. Those subsidizes could be expanded to at home offices for remote workers to carry out the same goal of sustaining productivity. If donors, sponsor t, and productivity of workers. Figure 3.4 shows an illustration of the concept.

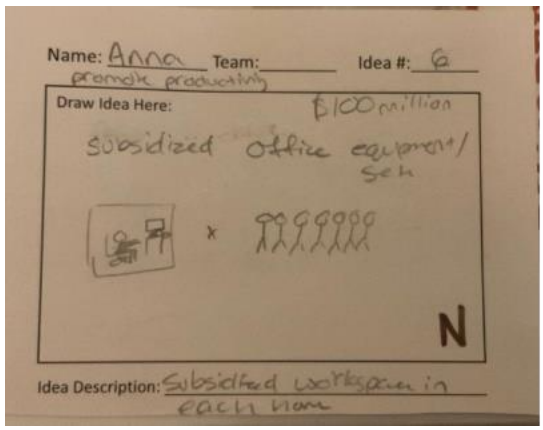

Figure 2.3.4. A sketch showing the subsidies for office equipment to increase comfort and productivity.

2.4 Sustainability-focused Ideation

During this set of ideations, us as a group focused on sustainability from carbon saving to personal life improvements. Sustainability is of utmost importance to this project because Penn States main goal and guiding principles are built on sustainability. To cut down on computer waist in Figure 2.4.1 we create a portable workstation made up of a laptop and supporting accessories. Figure 2.4.2 focuses on the energy uses of Penn State with adding solar panels the building. then in Figure 2.4.3 we focus on the employees' comfort with blue light style glasses. These all follow different parts of sustainability.

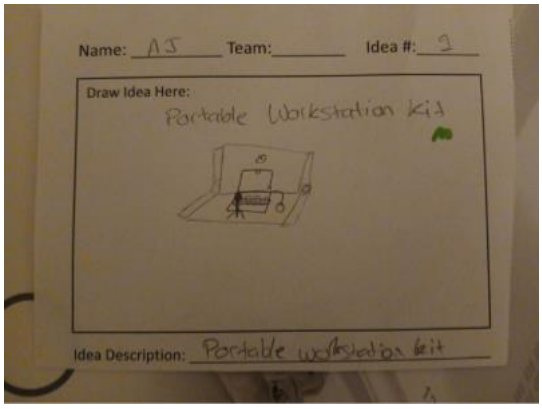

Figure 2.4.1. Laptop and supporting accessories in a briefcase, for a mobile workspace

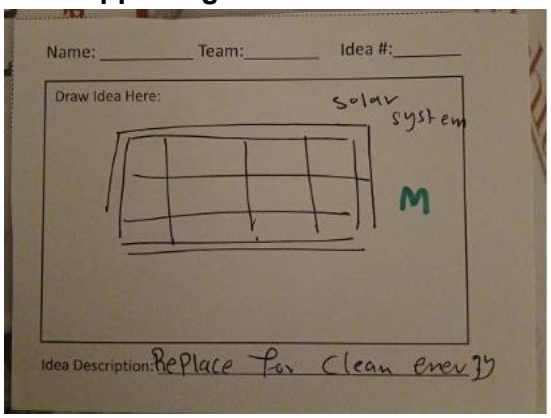

Figure 2.4.2. Array of solar panels for the top of the buildings

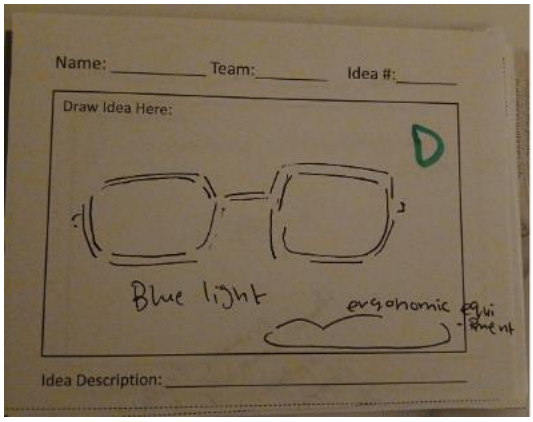

Figure 2.4.3. Set of blue light glasses to improve worker comfort

2.5 Preliminary Evaluation of Screened Concepts

Once we had completed the ideation portion of our design process, each team member chose one or two concepts to look at in-depth and consider for our final concept selection. Each teammate created a sketch of their idea and then used this to start a preliminary evaluation process. We used evaluation criteria based on both key design requirements as well as stakeholder needs. The four main criteria were “Promotes Productivity”, “Promotes Work/Life Balance”, “Reduce Greenhouse Gas Emissions”, and “Does Not Inhibit Robust University Workspace”. Each teammate evaluated their idea based on these four criteria and then chose two benchmarked products to compare against the same criteria. Through this evaluation, we gained a better sense of the strengths and weaknesses of our chosen ideas, as well as what pre-existing solutions are out there and inspiration we could gain from them.

2.5.1 Screened Concept 1

Screened concept 1 focused on creating a digital workspace that could accommodate hybrid employees, leading to better communication, collaboration, and overall employee productivity. As shown in **Figure 2.5.1** when employees signed in from their workstation, they would be able to access the status of the project they were working on, as well as seeing who had last contributed to the project. The last employee to edit a project would also be listed along with their contact information, so that an employee could ask questions about their work and overall have a smoother collaborative experience.

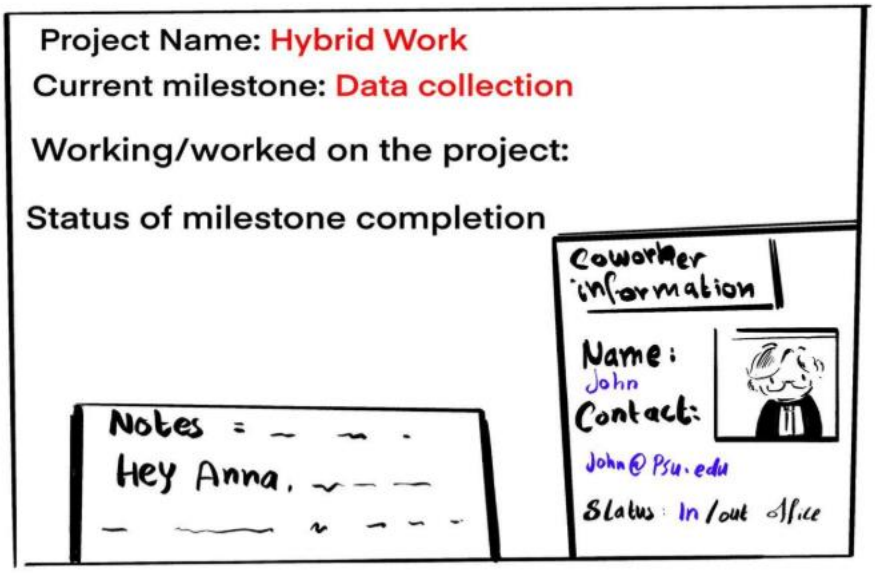

Figure 2.5.1. An illustration showing concept 1, the interface of the digital shared workspace.

The table below shows the screening of the concept 5a against existing products:

Table 3 (Part 1 out of 2): Screened concept 1 preliminary evaluation table against evaluation criteria and benchmarked products

Evaluation criteria	Preliminary evaluation of screened concept+ questions raised	Benchmarked product #1 Monday.com	Benchmarked product #2 Miro
Promote productivity	It helps with work distribution and makes employees more productive due to fairness and the ability to navigate the project better. Question: How is the work moved between coworkers without making the system rigid?	The website is great for promoting productivity, but it's extremely feature heavy and this has discouraged many users. People also mention that it lags when due to the same reason. It is also unnecessarily complicated for smaller teams. The website is ranked no.1 in the world in its field of trackers	The website covers the similar conceptually. It has sticky notes like the function of ms. Whiteboard. It promotes team thinking because of the collaborative features it offers. The interface is slightly less complicated than Monday.com, however some features are unnecessary. Distracting features might make work less productive.
Promote work/life balance	It helps knowing what's needed to be done and therefore helps employees get done in a good time and this reduces work stress, and the effective rotation helps with collaborative work	Monday.com has great milestone tracking features and it shows clear work distribution boundaries after work is assigned on it. One downside of Monday is that it is	Miro is good for group planning and notes sharing but not tracking. It might come in handy for small teams who are not documentation heavy. This makes it hard to achieve a

Table 3 (Part 2 out of 2): Screened concept 1 preliminary evaluation table against evaluation criteria and benchmarked products

	which promotes work/life balance Questions: How to encourage employees to take time to go on break or end of the day? Does that make it feature dense?	similar to an excel spreadsheet, it is a tracker, but it doesn't tell you when to stop working or when to take a break or when your workday is over.	work-life balance since tracking is not getting done by anyone. It is plainly a collaborative space with notes/files sharing and an ideation space.
Reduce greenhouse gas emissions	Less employees need to go into the office which reduces CO2 emission and can track their projects from home. Questions: What would work look like with the software onsite? What would employees do when website is down for maintenance?	Monday.com is a website. This means more internet usage on average and therefore more energy and more money spent. Making the concept into a software would be slightly better for the environment.	Miro might contribute by helping hybrid workers outside of the office see the notes and ideation from a far without the need of going into the office. This means that they would not need to commute to office for meetings or in office work passing. It reduces CO2 emissions.
Does not inhibit robust university workspace (i.e. liveliness)	No. This software helps making workers happier. Therefore, the work environment is going to be healthier and more effective. This is due to fairness and good attitude. Questions: Would there be a way to promote	No. It is a website that if used would promote productivity and organization within and outside of the office. It is a good tracker and is beneficial for bigger teams.	No. It has a positive user interface with lots of stickers and emojis and even GIFs. This is an advantage in the sense of brining humor to the office and brining familiarity between coworkers which helps maintain company culture and would help new hybrid employees
	socializing or would that be bloatware?		blend in easier with their team.

When evaluated with the key requirements in the preliminary evaluation process, this concept had a lot of strengths. It was a fully digital solution that encouraged hybrid work, which both promoted productivity between workers and was sustainable, as hybrid work reduces greenhouse gas emissions and overall creates a more environmentally friendly workplace. The evaluation table did raise some questions to help refine this concept and helped us to refine some important details (such as how this software might function if it undergoes maintenance, or how it could promote a better work / life balance with lunch breaks). This concept also performed better against some of the pre-

existing industry products, as it was a more streamlined solution with less distracting features, while still meeting the key requirements of our problem.

2.5.2 Screened Concept 2

Screened Concept 2 focused on creating a board that would help facilitate communication between workers both in-office and remotely. The idea, as shown in **Figure 2.5.1**, was to have a physical board in the office that would allow workers to indicate where they were working from for the day, as well as if they were absent for the day. Remote workers would be able to electronically access the board and use it to update their location. This would best be suited for workers in a smaller office space, where someone might want to ask their co-worker a question and could use it to see if they were in the office for the day.

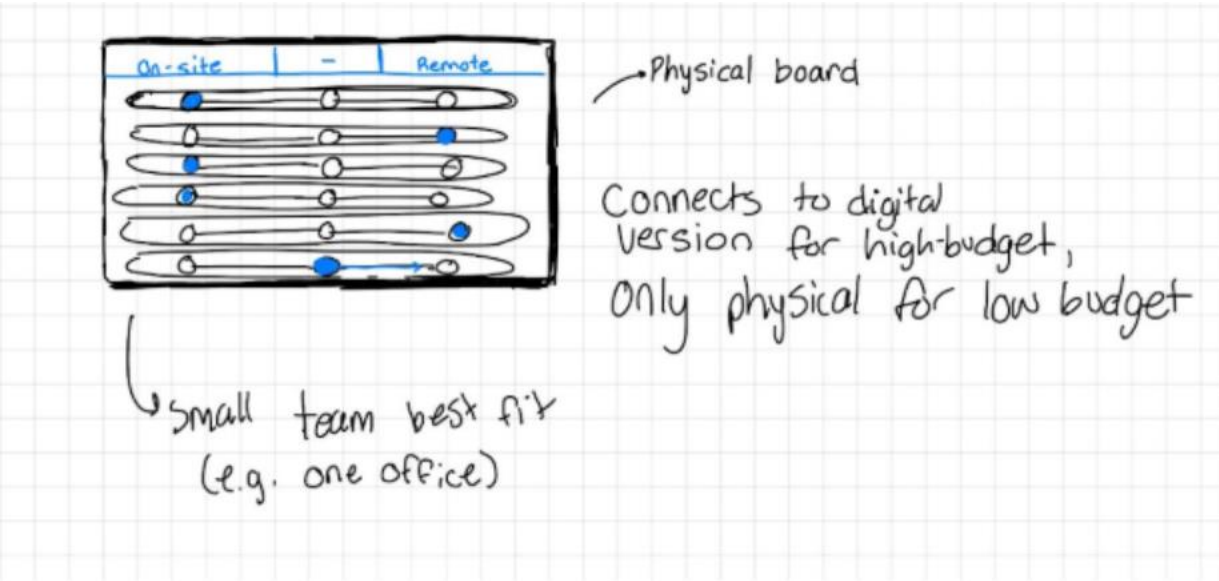

Figure 2.5.2. Sketch of screened concept 2, a board to indicate employee's location

Table 4: Screened concept 2 preliminary evaluation table, comparing against evaluation criteria and benchmarked products

	Screened Concept
Evaluation Criteria	**Preliminary Evaluation and Question Raised**
Promote Productivity	Better communication between co-workers --> better productivity in collaboration, facilitates quick communication and attendance accountability
Promote work / life balance	Allows easily for remote workers to still be involved, promoting remote work. What system would organize which workers are remote/on-site each day?
Reduce greenhouse gas emissions	Only requires a physical board in each office - limited resources. What might be the costs/carboon footprint of manufacturing the boards?
Does not inhibit robust university workspace	Better allows for visitors/students to know which staff they can contact in-person when visiting offices
	Benchmarked Product #1 - Connecteam
Promote Productivity	Involves multiple processes - task management, payroll, employee scheduling, etc. , which can greatly improve productivity if this is an existing issue. Led to 84% increase in scheduling speed according to website.
Promote work / life balance	Could be more difficult to use because of its many functions (e.g. not as simple for individual employees to use). Also potentially makes work-life balance difficult because tasks are always able to be viewed. Each employee will also see every update in the 'updates' tab which could overwhelm.
Reduce greenhouse gas emissions	Only a digital platform - less physical resources
Does not inhibit robust university workspace	Product is meant mainly for fully-digital teams, so does not as easily support hybrid communication. Additionally, students would not be able to access the information, making employees less accessible.
	Benchmarked Product #2 - Homebase
Promote Productivity	Helps to manage work schedules, timeclock (helpful for hourly employees), payroll, and other HR tasks. Would promote productivity and elimate miscommunications/confusion about scheduling and who is in office.
Promote work / life balance	Facilitates easy communication between employees and managers which could help to better facilitate work / life balance, but similar to Connecteam, could also lead to employees being overly accessible?
Reduce greenhouse gas emissions	Fully digital app which reduces greenhouse gas emissions as no physical product is needed.
Does not inhibit robust university workspace	Free app for employees but once again inaccessible information for students at university, making contacting employees in SEDI office less accessible.

When going through the preliminary evaluation process, this idea had multiple strengths that aligned with the key requirements. For example, by facilitating communication and allowing for managers to keep track of their staff, it also promoted productivity between workers. This concept is also a low-cost and low-resource solution, making it environmentally friendly and sustainable. Additionally, this would help university students who might want to stop in the office and ask a question to a staff member (by showing them who is available), which would create an overall more "robust university workspace". This concept was like some pre-existing products in the industry, but most of those products (including the examined benchmarked products) are fully digital, whereas this concept would be both a physical and electronic design.

2.5.3 Screened Concept 3

Screened Concept 3 focused on the daily workflow and creating a cohesive display that each worker could use, whether they were working from home or remotely. As shown in **Figure 2.5.3**, employees would log in with their company credentials and then be shown a display giving them valuable information for the day. For example, this might include which co-workers are online or present. This would also allow employees to easily contact each other and find each other's contact information. Similarly to the other ideas, this would facilitate hybrid communication which in turn would facilitate and promote employee productivity.

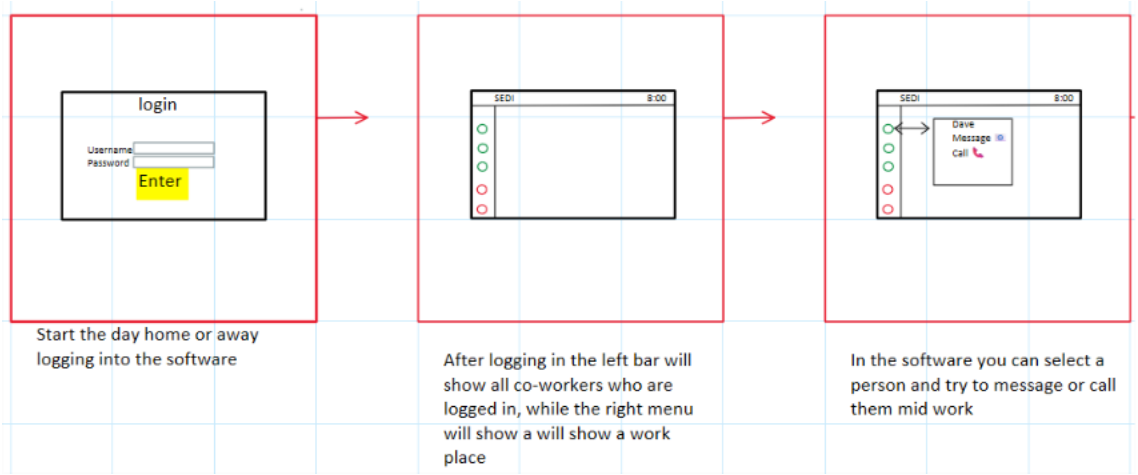

Figure 2.5.3. Sketch of screened concept 3 showcasing employee display and their expected workflow

When evaluating this concept against the key requirements and other stakeholder needs, this idea had a lot of strengths that could improve hybrid workflows. For example, this concept gave a clean and organized workspace, promoting better communication and therefore employee productivity. Being a fully digital design, it would reduce greenhouse gas emissions and promote hybrid workflows which would be an environmentally friendly solution. This solution would also promote a work / life balance as it improves employee productivity and facilitates employees to finish their work faster, while also not limiting them to strictly working in-person. When compared against other pre-existing products, this concept would be a more personalized and targeted solution for SEDI employees, which would allow more flexibility and overall promote better productivity because it could be suited to their exact projects and overall needs. It also would be a cyber-secure solution as it would be only for SEDI employees, creating more security for SEDI.

2.6 Concept Selection

This section should provide a discussion of your team's concept selection process, resulting in two unique concepts to explore with your users. Include your concept selection matrix along with a discussion of it. Make sure to discuss your criteria along with a discussion of how you arrived at your ratings. Clarly communicate the two concepts you are moving forward and why (note: you will describe the concepts in the following section).

To select our team's final product, we created a concept selection matrix that helped to both qualitatively and quantitatively analyze and select the best and most suitable product idea. Each team member's concept was rated against the key stakeholder needs of this project, and then given final scores based on the weighted importance of each stakeholder need (plus additional criteria that our group found important, which in this table, was "Fairness). Following the results in Table 5, we decided to move forward with Idea 1 and Idea 3, which was an Online Platform and ICWA (Integrated

Communication and Work Application). We found that these ideas were the best at meeting the key stakeholder needs while supporting hybrid work, and had similar features and goals that could be combined into one overall cohesive final product, which was PSU Orbit.

Table 5: Concept selection matrix, detailing each idea and their weighted scores against key criteria

		online platform		BEAM		ICWA	
Criteria (Needs)	Weights (%)	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
promotes productivity	18%	4	0.72	1	0.18	4	0.72
promotes work-life balance	22%	3	0.66	2	0.44	4	0.88
reduces emissions	12%	3	0.36	3	0.36	3	0.36
Supports robust university workspace	18%	3	0.54	3	0.54	3	0.54
efficient use of university resources	15%	4	0.6	1	0.15	3	0.45
Fairness	15%	5	0.75	4	0.6	4	0.6
Total	100%		3.63		2.27		3.55
Decision			Yes		No		Yes

3.0 Second Design Iteration

Product 1 Description: Online Platform

Product: A digital platform that connects remote and in office workers
Name of product: Online Platform
Mission: create a fair and productive work environment

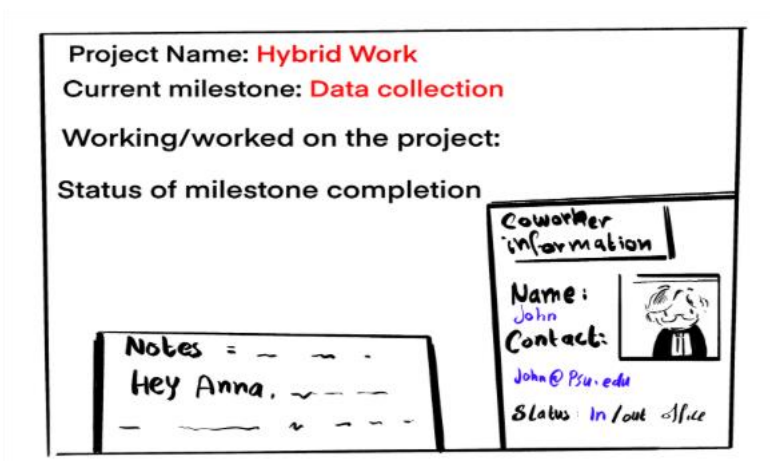

Figure 3.0.1. Product 1 Sketch detailing features of an online work platform

Features:

1. On the top: The software displays employees' names, tasks, and group projects shared. It would also include project milestones.

2. In the middle: Collaborative workspace for the workers

3. On the bottom: A space for notes and extra details
4. At the end of the day: Friendly reminders to stop working at the end of the day.

Product 2 Description: ICWA

Product: A digital workspace that increases team communication and employee productivity.

Name: ICWA – an Integrated Communication and Work Application

Misson: To improve productivity and collaboration in a hybrid environment

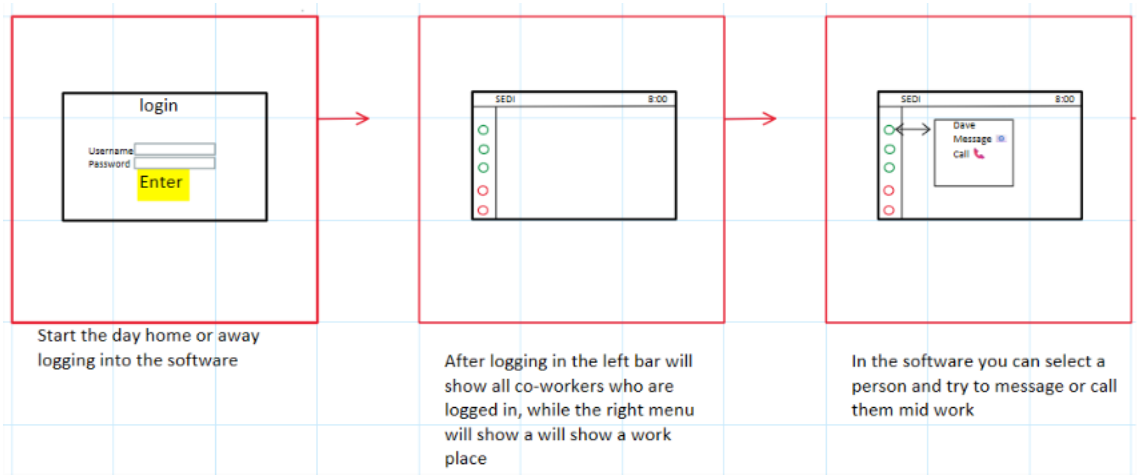

Figure 3.0.2. Sketch detailing features of Product 2, including log-in and overall workflow

Features:

On the left: A communication sidebar

a. This provides contact information and location of employees
b. Employees can contact co-workers (message, call) by tapping on their icon

On the right: A main workspace with necessary applications and software

c. This feature helps workers access the tools they need from any location, making productive hybrid work easier.

2. A flexible work cut-off feature
   a. ensuring employees take breaks and stop working at the end of workday
   b. Flexible: can extend the cutoff before the break.

3.1 User Interviews on Concept 1

After conducting user interviews on our previous interviewees, we found that the overall feedback for Concept 1 was that it would be a helpful addition to a hybrid workspace. The features that were most highly rated was its work-life balance features and its project organization features (specifically its project milestone and management options). The main feedback of aspects to improve were the details and execution of the work-life balance features so that it wouldn't unnecessarily inhibit the work of employees and allow for flexibility within their work schedule.

3.2 User Interviews on Concept 2

The overall feedback for Concept 2 was like Concept 1, although with a few differences in its main strengths and weaknesses. Like Concept 2, its features on project organization were a big strength, with its other large strengths being its security (with the log-in feature using PSU credentials), and its communication tools (with the left sidebar giving access to contact information and availability of co-workers). Its main feedback was that there could be more features included to support the work-life balance of workers, and a more detailed implementation of how existing productivity software would be implemented into ICWA itself.

3.3 Concept Selection

Describe your team's approach to selecting a revised idea, and include any tools used. Make sure to describe your new idea(s), highlighting how you refined it through additional ideation and discussion. Include key functions and/or components to demonstrate how your concept "works".

Our approach to selecting a final revised idea was to mainly combine the two previous ideas into a single, cohesive design that included the best features of both the previous designs. This was mainly created through collaborative discussion and sketching of what we wanted our final prototype to accomplish and look like, so that we could select the necessary features from our previous designs to include in our final prototype. Overall, essentially all the previous design features were included in our final design and then were refined to create a smoother and more user-friendly interface for SEDI. The final list of our features was:

1. Communication sidebar that shows contact information and status of co-workers
2. Main interface that holds necessary software for SEDI workers
3. Project organizer with milestone trackers and overall task details
4. Work-life balance features with reminders and questionnaires to support well-being.

4.0 Final Design and Evaluation

4.1 Final Design Description

The design of the final product was concluded after the second design iteration phase and the addition of edits to the concept. Our final prototype, a high-fidelity model was created using the web-based service, Figma [7].

The final product was given the name “PSU Orbit,” and is proposed as a solution to online work collaboration. This system is an online work platform that connects all employees and acts as a portal that can track work, projects, and tasks assigned.

This portal could also track whether a worker is scheduled as remote or in-office, while also announcing their online/away/offline status. It would also provide in-platform communication and other contact information under the worker’s profile, such as email addresses, work phone number, and other contact details.

Other features include providing detailed information about the project milestones for a group or displaying tasks assigned to an individual user. It also contains features that support the work-life balance of the employee, such as reminders at the end of the workday, and questionnaires to check-in on the employee’s well-being. The final product design is shown in **Figure 4.1.1,** with the main sections of the interface being:

1. The collaborative space
2. Tasks-assigned-to-workers’ space
3. Team space with all workers’ profiles
4. Project information

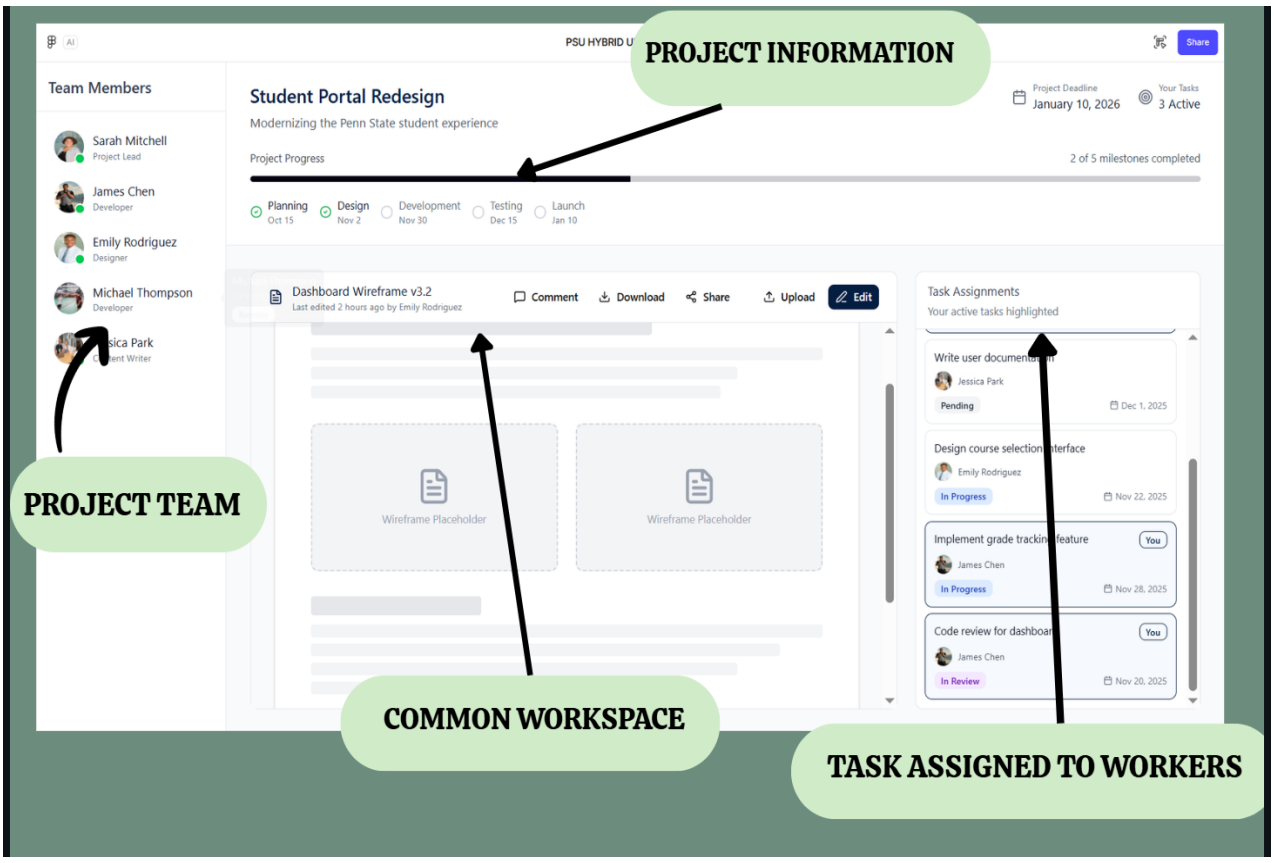

Fig 4.1.1. The final interface of the platform, along with arrows pointing to the main sections.

4.2 Design Evaluation

Design evaluation was an essential step in choosing and finalizing a concept. The evaluation starts with data collection, both quantitative and qualitative.

Our team used multiple surveys that covered different topics related to the concept of the product and hybrid work in general. Interviews were held with past and current hybrid workers inside and outside of SEDI. The qualitative assessment included raising different questions about different aspects of the design.

1. The quantitative assessment:

The surveys and metrics studied to understand the defects of the final product.

- **Surveys**

  Surveys online were a supplementary asset that aided our final product evaluation. The surveys covered topics such as hybrid employees' satisfaction with the level of fairness and communication in their offices.

- **Research**

  We also conducted preliminary research on hybrid work and specifically hybrid work within SEDI. This research was conducted independently by each group member, as well as collaboratively by analyzing the provided SEDI Statement of Work provided within the project resources. This provided important quantitative data points such as statistics about hybrid vs. in-person vs. remote work, as well as statistics about the number of hybrid workers and their overall rated satisfaction with a hybrid work system.

2. Qualitative assessment

- **Interviews**

The interviewees were from inside and outside of SEDI, and in different professions. This strengthened the usefulness of our qualitative results due to the interviewees' diverse experiences with hybrid work. The interviews from SEDI were prepared by our professor, while the interviews from outside of SEDI were done by our team.

<u>The methodology for the interviews:</u>

Two interviews were conducted at two different points in this project. The first was at the start of the problem understanding phase. The second was during the design evaluation process. Those two checkpoints were important for understanding our research and making sure our final design fits the needs of the workers. The first stage of interviews was helpful in gaining insights about what a hybrid work was like and the main benefits/problems that our interviewees had with a hybrid work system. This allowed us to better tailor our product and ideation to what those problems or "pain points" were, helping us to create an overall more effective and suitable product. The second stage

of interviews (conducted after our final idea was developed), was very helpful in providing feedback as to how an actual hybrid worker might use our product and served as checkpoint to help us gather input on things we need to change or additions to be made.

The questions during the interview revolved around the employee's perspective of hybrid work and the challenges they face. Other questions, such as “What they like about hybrid work,” were asked to gain a holistic view of their work-life. Notes were taken, and then interview results were shared, discussed, and then implemented into our product ideation, design, and refinement.

- **Comparative Benchmarking**

Another form of the qualitative data was gathered from customer reviews for different platforms that share similarities to our final product concept. The reviews were analyzed and chosen depending on consumer satisfaction levels. For example, some customers were unsatisfied due to slow learning curve and pricing of some platforms, while others were happier to have a solution for offsite collaboration. This was very helpful in refining our product as we were able to analyze what gaps in the market existed for hybrid work platforms, and the main complaints of users of these platforms that we could then address within our own product.

4.3 Sustainability

Sustainability has been a key focus of this product from the earliest stages of ideation. Additionally, Penn State has a large interest in promoting sustainability within their school and workforce, and so it was essential that we ensure our product has a positive impact on the environment and SEDI’s sustainability. Hybrid work comes with a unique impact on sustainability as it impacts the habits of workers and how they interact with the environment. Compared to a traditional in-person workplace, remote work reduces greenhouse gas emissions by cutting down on transport, building utility usage (e.g. gas, electricity), as well as other costs that come from having workers on-site. Our product facilitates hybrid work, making working from home a more realistic option for workers while maintaining the productivity of a fully in-person work environment. This means that building utility costs and transport-related greenhouse gas emissions will be reduced, improving SEDI’s sustainability.

4.4 Market Feasibility Analysis

The concept of creating a platform that acts as a collaborative space and connects works has been explored by different companies. Examples of those platforms include Monday.com, Miro, and Asana. Those platforms and online spaces are subscription-based and could work for the university’s staff. However, there are inherent benefits to adopting a private software domain. The first is the flexibility of updating the features of the software. Owning a private platform would allow for better control over data and privacy. Another benefit is the university’s control over the software's policy.

The cost analysis was done by comparing the prices of similar market products and creating an estimate that was found to be between a minimum of $300k and a maximum of $1 million, with an

average of repairs ranging between $75k and $250k, depending on the quality of implementation of the platform.

Overall, having a private platform tailored to fit the university's staff would be a better long-term investment than paying for subscription models. An additional step that could be taken to maximize the returns on this investment is to utilize the platform for multiple departments across Penn State. This would justify the investment even further.

5.0 Conclusions and Recommendations

Our final product is a online digital platform interface that facilitates hybrid work collaboration by connecting co-workers and providing easy access to communication and productivity tools, with specific features to ensure a healthy work-life balance. We created this product idea by going through each step of the design process: Empathy, Define, Ideate, Prototype, and Test. At each step, we identified and considered the key stakeholders and their requirements for this product to design a final solution catered to Penn State and SEDI's workforce and overall goals. As such, these stakeholder needs are specifically designed for and addressed in our final product. Our final product first and foremost promotes productivity within hybrid work, which was the main goal of this project. However, we also specifically designed for the other key requirements – our final product has features that intentionally promote a work-life balance (e.g. reminders), while also inherently promoting a balance by facilitating hybrid work. Greenhouse gas emissions are reduced by reducing the need for in-person commuting, and throughout all of this, the "robust university workspace" is maintained through creating an accessible way to include remote employees in on-site communication and work.

Although our final product is quite suitable for end-user of SEDI and Penn State, if a next version of this project was created, we would likely want to iterate more on the features of our platform to further promote the productivity of employees. This might look like additional communication features, refinement of the project/milestone trackers depending on user feedback, or other changes based on the feedback of actual SEDI users and ways to support their work further.

Overall, however, we believe that our product is a simple yet effective way to support hybrid work within SEDI while reducing the pain points of a hybrid workforce, making it a suitable and effective solution.

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Appendix A: (Label as A, B, C, D...)

<Start the each appendix section on a separate page. Material that is too extensive to include in the main text can be included in an appendix. Otherwise, delete.>